5. Carathéodory measurability and the coincidence criterion for measures

REAL VARIABLES I

Fall 2026

An outer measure μ^* is defined on every subset of a set X, however, according to Vitali's counterexample, the identity

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$$

may fail even though $A \cap B = \emptyset$. Measures, on the other hand, are defined on σ-algebras, which are mostly proper subsets of 2^X, are functionals for which the identity above (and its countable analogue) holds by definition. We will show how to find a "maximal" σ-algebra $\mathcal{M}$ associated to μ^* such that $\mu^*|_{\mathcal{M}}$ is disjointly additive and therefore a measure. We will also show that for the Lebesgue outer measure, the resulting σ-algebra $\mathcal{M}_{\mathcal{L}^*}$ contains all Borel sets.
Finally, we will prove a uniqueness principle: two measures that coincide on a suitable generating family coincide on the generated σ-algebra. This will allow us to prove that on the Borel σ-algebra, the Lebesgue meausure is indeed the unique one that assigns to intervals (or axes-parallel rectangles) their Euclidean volumes.
After studying these notes, you will be able to do the following.

- Apply the Carathéodory criterion to contstruct σ-algebras.
- Construct a complete measure by restricting an outer measure to its Carathéodory measurable sets.
- Show that the Lebesgue measure is a measure on Borel sets.
- Use Dynkin's π-λ theorem to prove that finite or σ-finite measures coincide on a generated σ-algebra under mild conditions.

5.1 The Carathéodory criterion

Throughout this section, X is a set and $\mu^* \colon \mathcal{P}(X) \to [0, \infty]$ is an outer measure. The central idea for defining the set A to be measurable is to require that one can "cut" *any* set $E \subset X$ using A by writing $E = (E \cap A) \cup (E \cap A^c)$. Given that the the two sets on the right hand side are disjoint, we require that $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A)$.

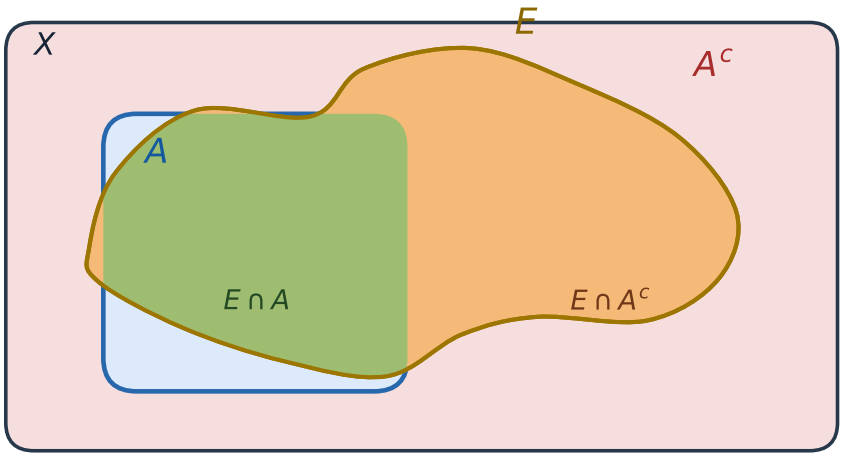


Figure 1: Within X, blue represents A, red represents A^c, and yellow represents E. Their overlap is green in $E \cap A$ and orange in $E \cap A^c$.

Figure description: Inside one large rounded rectangle labelled X, a blue rounded rectangle labelled A leaves space around it in red A complement. A yellow irregular set E crosses the boundary of A. The colors blend to green in E intersection A and orange in E intersection A complement.

Definition 1 (Carathéodory measurable set). A set $A \subseteq X$ is *Carathéodory measurable with respect to* μ^* if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \qquad \text{for every } E \subseteq X. \tag{1}$$

The collection of all such sets is denoted by $\mathcal{M}_{\mu^*}$.

The above defintion imposes no restriction on the set $E \subseteq X$ being cut. Note that subadditivity of μ^* always gives $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \setminus A)$, therefore, the definition above amounts to requiring that $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$.
It is immediate to check that $\emptyset$ and X are always Carathéodory measurable. Our goal is to show that:

- $\mathcal{M}_{\mu^*}$ is a σ algebra;
- the outer measure $\mathcal{M}_{\mu^*}$ is finitely additive on disjoint sets in $\mathcal{M}_{\mu^*}$.

That way, restricting μ^* to $\mathcal{M}_{\mu^*}$ we obtain a measure.
We begin by showing that $\mathcal{M}_{\mu^*}$ is an algebra

Proposition 2 (Finite unions and complements). The collection $\mathcal{M}_{\mu^*}$ is an algebra i.e. it is closed under finite unions and complements. Therefore it is also closed under finite intersections and set differences. Finally, since $\emptyset \in \mathcal{M}_{\mu^*}$, it follows that $\mathcal{M}_{\mu^*}$ is an algebra.

Proof of Proposition 2. Step 1: prove closure under finite unions.
Let $A, B \in \mathcal{M}_{\mu^*}$ and fix $E \subseteq X$. The diagram in Figure 2 shows the four pieces obtained by making both cuts.

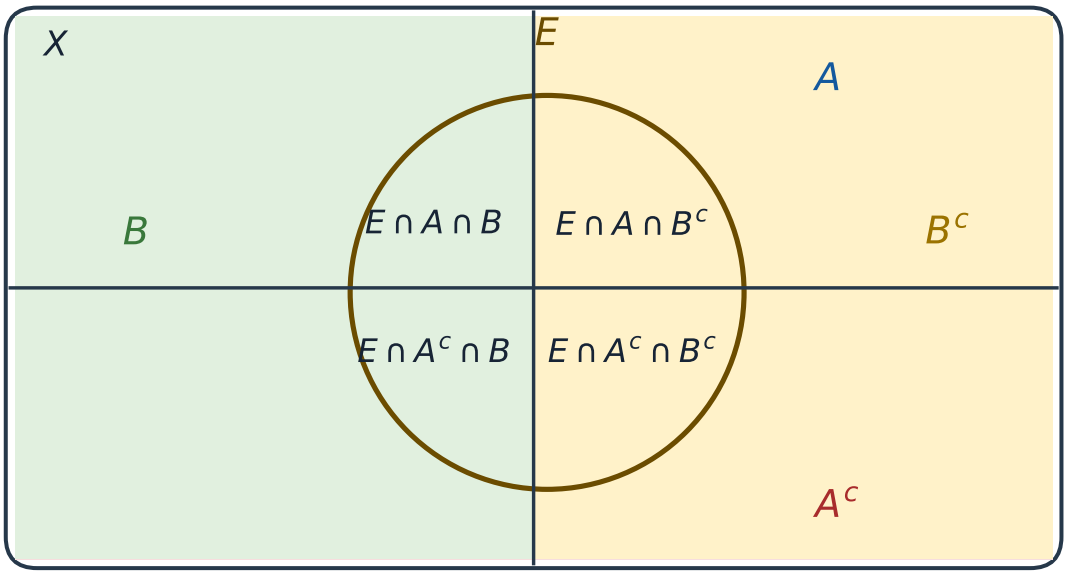


Figure 2: The horizontal cut by A and A^c, followed by the vertical cut by B and B^c, divides E into four pieces.

Figure description: Inside a rectangle labelled X, a horizontal line divides A above from A complement below, and a vertical line divides B on the left from B complement on the right. A central circle labelled E is quartered and its four pieces are labelled E intersection A intersection B, E intersection A intersection B complement, E intersection A complement intersection B, and E intersection A complement intersection B complement.

The two cuts give

$$\begin{aligned}\mu^*(E) &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &\geq \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c).\end{aligned}$$

The first inequality holds by measurability of A. The second holds by measurability of B, applied to $E \cap A$ and $E \cap A^c$, respectively. Since the first three pieces cover $E \cap (A \cup B)$, subadditivity gives

$$\begin{aligned}&\mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) \\ &\qquad \geq \mu^*(E \cap (A \cup B)).\end{aligned}$$

The fourth piece is $E \cap (A \cup B)^c$. Combining this inequality with the preceding lower bound gives

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c).$$

The reverse inequality follows from subadditivity, so $A \cup B$ satisfies the Carathéodory criterion. Induction gives closure under finite unions.

Step 2: prove closure under complements.

The defining equality is symmetric in A and A^c, because $E \cap A^c$ is the part of E outside A. Thus $A \in \mathcal{M}_{\mu^*}$ implies $A^c \in \mathcal{M}_{\mu^*}$.

Step 3: recover intersections and differences.

De Morgan's laws give closure under finite intersections, and $A \cap B^c$ gives closure under differences. □

Next, we show that μ^* if finitely disjointly additive on $\mathcal{M}_{\mu^*}$.

Proposition 3 (Finite additivity with a measurable union). Let A, B. If $A \cap B = \emptyset$, then $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$.

Proof of Proposition 3. Apply the Carathéodory criterion for A to the test set $A \cup B$. Since $A \cap B = \emptyset$, it gives

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \setminus A) = \mu^*(A) + \mu^*(B).$$

□

Since μ^* is finitely disjointly additive it is σ disjointly superaddtive. Being an outer measure μ^* is σ-subadditive. Therefore to show that μ^* is a measure on $\mathcal{M}_{\mu^*}$ it is sufficient to show that $\mathcal{M}_{\mu^*}$ is a σ algebra. This is indeed the content of the next statement.

Proposition 4 (Countable unions of measurable sets). The collection $\mathcal{M}_{\mu^*}$ is closed under countable unions. Consequently, it is a σ-algebra.

Proof of Proposition 4. Step 1: treat pairwise disjoint unions.
Let $(A_n \in \mathcal{M}_{\mu^*})_{n \geq 1}$ be pairwise disjoint and set $A = \bigcup_{n=1}^{\infty} A_n$. Fix $E \subseteq X$. Since $\mathcal{M}_{\mu^*}$ is closed with respect to finite unions we have

$$\mu^*(E) = \mu^*(E \cap \bigcup_{n=1}^{N} A_n) + \mu^* \left(E \setminus \bigcup_{n=1}^{N} A_n \right).$$

By finite additivity of μ^* on $\mathcal{M}_{\mu^*}$ we obtain that

$$\mu^*(E) = \sum_{n=1}^{N} \mu^*(E \cap A_n) + \mu^* \left(E \setminus \bigcup_{n=1}^{N} A_n \right)$$

This step is requires justification because $E \cap A_n$ may fail to be in $\mathcal{M}_{\mu^*}$. However, it holds that

$$\begin{aligned} \mu^*(E \cap \bigcup_{n=1}^{N} A_n) &= \mu^* \Big((E \cap \bigcup_{n=1}^{N} A_n) \cap A_N \Big) + \mu^* \Big((E \cap \bigcup_{n=1}^{N} A_n) \cap A_N^c \Big) \\ &= \mu^* \Big(E \cap A_N \Big) + \mu^* \Big(E \cap \bigcup_{n=1}^{N-1} A_n \Big). \end{aligned}$$

Applying this procedure N times yields the desired identity $\sum_{n=1}^{N} \mu^*(E \cap A_n) = \mu^*(E \cap \bigcup_{n=1}^{N} A_n)$. Next, by monotonicity, we get that

$$\mu^*(E) \geq \sum_{n=1}^{N} \mu^*(E \cap A_n) + \mu^*(E \setminus A)$$

for every $N \in \mathbb{N}$. Taking the supremum over N, gives

$$\mu^*(E) \geq \sum_{n=1}^{\infty} \mu^*(E \cap A_n) + \mu^*(E \setminus A).$$

Finally, applying σ subadditivity of μ^* to $E \cap A = \bigcup_{n=1}^{\infty} (E \cap A_n)$, gives

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \setminus A).$$

The reverse inequality follows from subadditivity. Thus $A \in \mathcal{M}_{\mu^*}$.

Step 2: disjointify an arbitrary countable union.
Let $(B_n \in \mathcal{M}_{\mu^*})_{n\geq 1}$ be arbitrary. Define

$$D_1 = B_1, \qquad D_n = B_n \setminus \bigcup_{k=1}^{n-1} B_k \quad (n \geq 2).$$

By Proposition 2, every $D_n \in \mathcal{M}_{\mu^*}$. The sets D_n are pairwise disjoint and $\bigcup_{n=1}^{\infty} D_n = \bigcup_{n=1}^{\infty} B_n$, so their union (and hence $\bigcup_{n=1}^{\infty} B_n$) is measurable by Step 1.

□

5.1.1 Completeness of measures

Let $\mathcal{A}$ be a σ-algebra on X, and let ν be a measure on $\mathcal{A}$. We say that $\mathcal{A}$ is *complete with respect to* ν if for any $N \in \mathcal{A}$ with $\nu(N) = 0$ and any $M \subset N$ one has $M \in \mathcal{A}$. An equivalent characterization is that if $A \in \mathcal{A}$ and $B \subset X$ such that $A \Delta B \subset N$ for some $N \in \mathcal{A}$ with $\nu(N) = 0$ then $B \in \mathcal{A}$.

Corollary 5 (Carathéodory construction). The restriction $\mu = \mu^*|_{\mathcal{M}_{\mu^*}} : \mathcal{M}_{\mu^*} \to [0, \infty]$ is a measure.

The preceding propositions establish the required facts. They show that $\mathcal{M}_{\mu^*}$ is a σ-algebra and that μ^* is finitely additive on its pairwise disjoint measurable sets. The result from the previous notes therefore gives σ-superadditivity on pairwise disjoint families. Together with the σ-subadditivity of the outer measure, this gives σ-additivity of the restriction.

Proposition 6 (Completeness of the Carathéodory σ-algebra). The σ-algebra $\mathcal{M}_{\mu^*}$ is complete with respect to the measure $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$.

Proof of Proposition 6. Exercise. □

5.2 Lebesgue measurable sets

Fix $d \geq 1$. Let $\mathcal{L}_d^*$ denote Lebesgue outer measure on $\mathbb{R}^d$, generated by countable covers by bounded open axis-parallel rectangles and their d-dimensional volumes. We next verify that the Carathéodory domain of $\mathcal{L}_d^*$ contains the Borel σ-algebra.

5.2.1 Open rectangles and Borel sets

The open rectangles used to define $\mathcal{L}_d^*$ are already Carathéodory measurable. The point is that the part of a covering rectangle outside a fixed open rectangle can be covered by finitely many open rectangles, with an arbitrarily small increase in total volume.

Proposition 7 (Bounded open rectangles are measurable). Every bounded open axis-parallel rectangle in $\mathbb{R}^d$ is Carathéodory measurable with respect to $\mathcal{L}_d^*$.

Proof of Proposition 7. Let R be a bounded open axis-parallel rectangle. Fix $E \subseteq \mathbb{R}^d$. Subadditivity gives

$$\mathcal{L}_d^*(E) \leq \mathcal{L}_d^*(E \cap R) + \mathcal{L}_d^*(E \cap R^c).$$

We prove the reverse inequality. It is automatic when $\mathcal{L}_d^*(E) = \infty$, so assume $\mathcal{L}_d^*(E) < \infty$, and fix $\varepsilon > 0$.

Step 1: choose an open-rectangle cover of E.

Choose bounded open axis-parallel rectangles $(Q_n)_{n\geq 1}$ that cover E and satisfy

$$\sum_{n=1}^{\infty} \mathrm{vol}_d(Q_n) \leq \mathcal{L}_d^*(E) + \frac{\varepsilon}{2}.$$

For each n, the intersection

$$U_n = Q_n \cap R$$

is an open rectangle (or is empty), and the sets U_n cover $E \cap R$.

Step 2: cover the complementary part of each Q_n.

For every n, the set $Q_n \cap R^c$ has an open-rectangle cover $V_{n,1}, \ldots, V_{n,m_n}$, where $m_n \leq 3^d$, such that

$$\sum_{r=1}^{m_n} \mathrm{vol}_d(V_{n,r}) \leq \mathrm{vol}_d(Q_n) - \mathrm{vol}_d(U_n) + \varepsilon 2^{-n-1}.$$

The figure illustrates this elementary geometric fact in dimensions one and two.

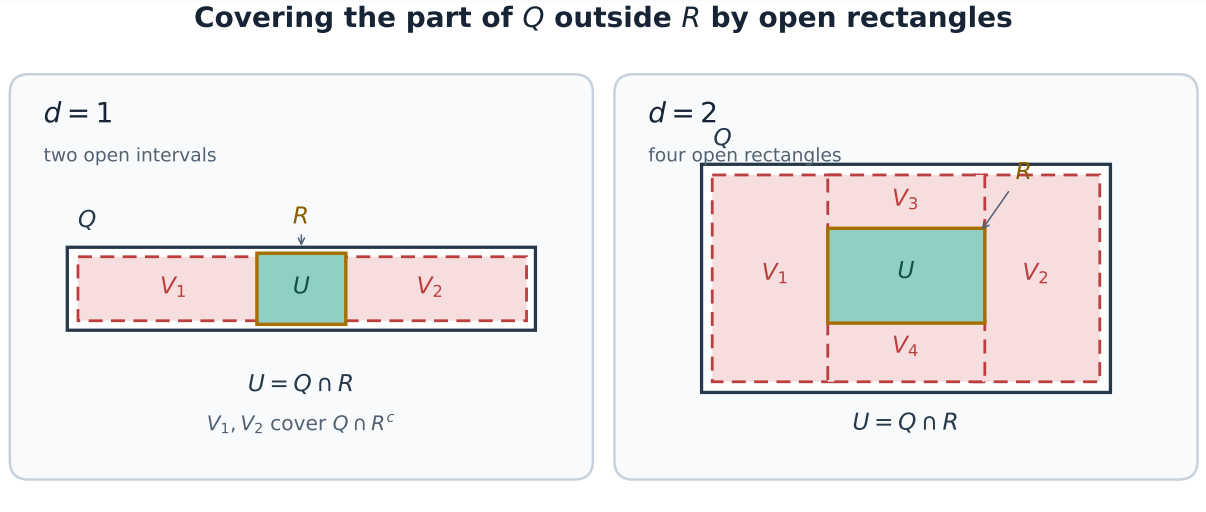


Figure 3: The part of an open rectangle Q outside an open rectangle R can be covered by at most 3^d open rectangles V_r, while $U = Q \cap R$ is itself open. Dashed boundaries indicate the arbitrarily small enlargement used to keep every covering rectangle open.

Figure description: Two diagrams. In dimension one, an interval Q contains an interval R; their intersection U is the middle portion, and the two remaining portions are covered by slightly enlarged open intervals V one and V two. In dimension two, a rectangle Q contains a central rectangle R; their intersection U is central, and the surrounding four regions are covered by slightly enlarged open rectangles V one through V four.

For $d = 1$, the complement of an open interval inside Q_n consists of its left and right portions. Enlarging each endpoint slightly gives two open intervals whose total length is within $\varepsilon 2^{-n-1}$ of the length of the complementary part. For $d = 2$, use the left and right strips of Q_n outside R, followed by the lower and upper strips in the middle; after slightly enlarging their boundaries, these are four open rectangles. Their total area can be made within $\varepsilon 2^{-n-1}$ of the area of the complementary part. In general, divide each coordinate direction into its left, middle, and right parts relative to R. The resulting grid has at most 3^d rectangular pieces, and every piece except the middle one lies outside R. Slightly enlarging those pieces gives the stated open-rectangle cover.

The rectangles $V_{n,r}$ cover $E \cap R^c$. Hence

$$\begin{aligned}\mathcal{L}_d^*(E \cap R) + \mathcal{L}_d^*(E \cap R^c) &\leq \sum_{n=1}^{\infty} \left(\mathrm{vol}_d(U_n) + \sum_{r=1}^{m_n} \mathrm{vol}_d(V_{n,r}) \right) \\ &\leq \sum_{n=1}^{\infty} \mathrm{vol}_d(Q_n) + \sum_{n=1}^{\infty} \varepsilon 2^{-n-1} \\ &\leq \mathcal{L}_d^*(E) + \varepsilon.\end{aligned}$$

Letting $\varepsilon \downarrow 0$ proves the reverse inequality, so R is measurable. □

Corollary 8 (Borel sets are Carathéodory measurable). Every Borel subset of $\mathbb{R}^d$ is Carathéodory measurable with respect to $\mathcal{L}_d^*$. Indeed, open axis-parallel rectangles generated $\mathcal{B}(\mathbb{R}^d)$, whereas $\mathcal{M}_{\mathcal{L}_d^*}$ is a σ-algebra containing such rectangles.

5.3 The coincidence criterion for measures

The Carathéodory theorem constructs a measure from an outer measure, thereby proving the existence of a measure that extends a cost under mild assumptions (σ-subadditivity).
Next, we investigate uniqueness: suppose two measures are defined on the same σ-algebra, and their values are known to agree on a generating collection. When can we claim that they agree on their entire domain of definition?

Definition 9 (π-system and Dynkin system). Let X be a set.

π-system A collection $\mathcal{P} \subseteq \mathcal{P}(X)$ is a *π-system* if $A \cap B \in \mathcal{P}$ whenever $A, B \in \mathcal{P}$.

λ-system A collection $\mathcal{D} \subseteq \mathcal{P}(X)$ is a *λ-system*, if

full set $\emptyset, X \in \mathcal{D}$;

complement $A \in \mathcal{D}$ implies $A^c \in \mathcal{D}$

sjoint countable union $\bigcup_{n=1}^{\infty} A_n \in \mathcal{D}$ for every pairwise disjoint sequence $(A_n \in \mathcal{D})_{n \geq 1}$.

The smallest λ system containing a collection $\mathcal{A}$ is denoted by $\lambda(\mathcal{A})$.

The intersection of any family of λ systems is a λ system, and 2^X is a λ system. Thus $\lambda(\mathcal{A})$ exists and coincides with the intersection of all λ systems containing $\mathcal{A}$.

Lemma 10 (Difference closure in a λ system). If $\mathcal{D}$ is a λ system and $A, B \in \mathcal{D}$ satisfy $A \subseteq B$, then $B \setminus A \in \mathcal{D}$.

Proof of Lemma 10. The sets A and B^c are disjoint members of $\mathcal{D}$. After adjoining empty sets, closure under disjoint countable unions gives $A \cup B^c \in \mathcal{D}$. Closure under complements gives

$$B \setminus A = (A \cup B^c)^c \in \mathcal{D}.$$

□

Theorem 11 (Dynkin's π-λ theorem). If $\mathcal{P}$ is a π-system on X, then

$$\lambda(\mathcal{P}) = \sigma(\mathcal{P}).$$

Proof of Theorem 11. The proof promotes intersection closure first from $\mathcal{P}$ to $\mathcal{P} \times \mathcal{D}$, and then from there to $\mathcal{D} \times \mathcal{D}$.

Step 1: reduce the theorem to intersection closure.

Set $\mathcal{D} = \lambda(\mathcal{P})$. Since every σ-algebra is a Dynkin system,

$$\mathcal{D} \subseteq \sigma(\mathcal{P}).$$

It remains to prove that $\mathcal{D}$ is a σ-algebra. For this, it is enough to prove that $\mathcal{D}$ is closed under finite intersections: such closure, together with closure under complements, gives closure under finite unions. If $(A_n)_{n\geq 1}$ is any sequence in $\mathcal{D}$, set

$$B_1 = A_1, \qquad B_n = A_n \setminus \bigcup_{j=1}^{n-1} A_j \quad (n \geq 2).$$

Then $B_n \in \mathcal{D}$, the sets B_n are pairwise disjoint, and $\bigcup_{n\geq 1} A_n = \bigcup_{n\geq 1} B_n$. Closure under disjoint countable unions therefore gives $\bigcup_{n\geq 1} A_n \in \mathcal{D}$, proving that $\mathcal{D}$ is a σ-algebra once finite-intersection closure is established.

Step 2: intersect an arbitrary Dynkin member with a generator.

Fix $A \in \mathcal{P}$ and define

$$\mathcal{D}_A = \{B \in \mathcal{D} : A \cap B \in \mathcal{D}\}.$$

This is a Dynkin system. Indeed, $A \cap X = A \in \mathcal{D}$. If $B \in \mathcal{D}_A$, then $A \cap B \subseteq A$, and Lemma 10 gives

$$A \cap B^c = A \setminus (A \cap B) \in \mathcal{D}.$$

If the sets $B_n \in \mathcal{D}_A$ are pairwise disjoint, then the sets $A \cap B_n$ are pairwise disjoint members of $\mathcal{D}$, and hence

$$A \cap \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A \cap B_n) \in \mathcal{D}.$$

Thus $\mathcal{D}_A$ is a Dynkin system. Since $\mathcal{P}$ is a π-system, $A \cap B \in \mathcal{P} \subseteq \mathcal{D}$ for every $B \in \mathcal{P}$. Therefore $\mathcal{P} \subseteq \mathcal{D}_A$, and minimality of $\mathcal{D}$ gives $\mathcal{D}_A = \mathcal{D}$. We have proved

$$A \cap B \in \mathcal{D} \qquad \text{whenever } A \in \mathcal{P} \text{ and } B \in \mathcal{D}. \tag{2}$$

Step 3: extend intersection closure to two Dynkin members.

Now fix $B \in \mathcal{D}$ and define

$$\mathcal{E}_B = \{A \in \mathcal{D} : A \cap B \in \mathcal{D}\}.$$

The same complement, proper-difference, and disjoint-union argument shows that $\mathcal{E}_B$ is a Dynkin system. By (2), it contains $\mathcal{P}$. Hence $\mathcal{E}_B = \mathcal{D}$. Since $B \in \mathcal{D}$ was arbitrary, $\mathcal{D}$ is closed under finite intersections. Thus it is a σ-algebra, so $\sigma(\mathcal{P}) \subseteq \mathcal{D}$. □

5.3.1 Finite and sigma-finite measures

Theorem 12 (Coincidence criterion for finite measures). Let $\mathcal{P}$ be a π-system on X, and let $\mathcal{A} = \sigma(\mathcal{P})$. Suppose μ and ν are finite measures on $(X, \mathcal{A})$ such that

$$\mu(X) = \nu(X) \qquad \text{and} \qquad \mu(P) = \nu(P) \quad \text{for every } P \in \mathcal{P}.$$

Then $\mu(A) = \nu(A)$ for every $A \in \mathcal{A}$.

Proof of Theorem 12. Consider the coincidence class

$$\mathcal{D} = \{A \in \mathcal{A} : \mu(A) = \nu(A)\}.$$

The equality of the total masses gives $X \in \mathcal{D}$. If $A \in \mathcal{D}$, finiteness gives

$$\mu(A^c) = \mu(X) - \mu(A) = \nu(X) - \nu(A) = \nu(A^c),$$

so $A^c \in \mathcal{D}$. If $(A_n \in \mathcal{D})_{n \geq 1}$ is pairwise disjoint, countable additivity gives

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} \nu(A_n) = \nu\left(\bigcup_{n=1}^{\infty} A_n\right).$$

Thus $\mathcal{D}$ is a Dynkin system. It contains $\mathcal{P}$, so Theorem 11 gives $\mathcal{A} = \sigma(\mathcal{P}) \subseteq \mathcal{D}$. □

Remark (When the generator contains the whole space). If $X \in \mathcal{P}$, agreement on $\mathcal{P}$ already includes the hypothesis $\mu(X) = \nu(X)$. The separate total-mass hypothesis is needed when $X \notin \mathcal{P}$.

For measures with infinite total mass, apply the finite result on an increasing exhaustion.

Theorem 13 (Coincidence criterion for sigma-finite measures). Let $\mathcal{P}$ be a π-system on X, let $\mathcal{A} = \sigma(\mathcal{P})$, and let μ, ν be measures on $(X, \mathcal{A})$. Assume that:

1. $\mu(P) = \nu(P)$ for every $P \in \mathcal{P}$;
2. there is an increasing sequence $(P_n \in \mathcal{P})_{n \geq 1}$ such that

$$X = \bigcup_{n=1}^{\infty} P_n \qquad \text{and} \qquad \mu(P_n) = \nu(P_n) < \infty \quad \text{for every } n.$$

Then $\mu(A) = \nu(A)$ for every $A \in \mathcal{A}$.

Proof of Theorem 13. Exercise □

Warning (Finiteness cannot be omitted). Agreement on a generating π-system does not by itself imply agreement on the generated σ-algebra. On $\mathbb{N}$, let

$$T_n = \{n, n+1, n+2, \ldots\}, \qquad \mathcal{P} = \{T_n : n \in \mathbb{N}\}.$$

The family $\mathcal{P}$ is a π-system and generates $\mathcal{P}(\mathbb{N})$, since $\{n\} = T_n \setminus T_{n+1}$ and every subset of $\mathbb{N}$ is a

countable union of singletons. Counting measure μ and the measure

$$\nu(E) = \begin{cases} 0, & E = \emptyset, \\ \infty, & E \neq \emptyset \end{cases}$$

are countably additive: for ν, both sides of the countable-additivity identity are 0 when every set is empty and are ∞ otherwise. They agree on every T_n, but $\mu(\{n\}) = 1 \neq \infty = \nu(\{n\})$. The measure ν has no countable exhaustion by sets of finite ν-measure.

5.3.2 Application to Borel measures

Theorem 14 (Coincidence on half-open rectangles). Let μ and ν be Borel measures on $\mathbb{R}^d$ that are finite on compact sets. Suppose

$$\mu\left(\prod_{j=1}^{d}(a_j, b_j]\right) = \nu\left(\prod_{j=1}^{d}(a_j, b_j]\right)$$

whenever $-\infty < a_j < b_j < \infty$ for every j. Then $\mu = \nu$ on $\mathcal{B}(\mathbb{R}^d)$.

Proof of Theorem 14. The proof verifies the three inputs of the σ-finite coincidence criterion: a π-system, generation, and a finite exhaustion.

Step 1: verify the π-system property.

Let $\mathcal{P}$ consist of $\emptyset$ and all bounded half-open rectangles $\prod_{j=1}^{d}(a_j, b_j]$. Intersections of such rectangles are again half-open rectangles or empty, so $\mathcal{P}$ is a π-system.

Step 2: show that the half-open rectangles generate the Borel sets.

Every member of $\mathcal{P}$ is Borel. Conversely, if $R = \prod_{j=1}^{d}(a_j, b_j)$ is an open rectangle, then R is the union of all half-open rectangles

$$\prod_{j=1}^{d}(p_j, q_j] \quad \text{with rational endpoints and} \quad a_j < p_j < q_j < b_j.$$

This is a countable union. Thus open rectangles, and hence all open sets, belong to $\sigma(\mathcal{P})$. Therefore $\sigma(\mathcal{P}) = \mathcal{B}(\mathbb{R}^d)$.

Step 3: construct a common finite exhaustion.

The rectangles

$$P_n = (-n, n]^d, \qquad n \geq 1,$$

form an increasing sequence in $\mathcal{P}$ whose union is $\mathbb{R}^d$. Each P_n is contained in the compact cube $[-n, n]^d$. Finiteness on compact sets and monotonicity give

$$\mu(P_n) = \nu(P_n) < \infty.$$

The result now follows from Theorem 13. □

Corollary 15 (Uniqueness of Lebesgue measure). Lebesgue measure on the Borel subsets of $\mathbb{R}^d$ is the unique Borel measure that is finite on compact sets and assigns every bounded half-open rectangle its d-dimensional volume. Its completion is the measure λ_d on $\mathcal{L}(\mathbb{R}^d)$.